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Modeling of High-Frequency Seismic-Wave Scattering and Propagation Using Radiative Transfer Theory

by Yuehua Zeng

Abstract This is a study of the nonisotropic scattering process based on radiative transfer theory and its application to the observation of the **M** 4.3 aftershock recording of the 2008 Wells earthquake sequence in Nevada. Given a wide range of recording distances from 29 to 320 km, the data provide a unique opportunity to discriminate scattering models based on their distance-dependent behaviors. First, we develop a stable numerical procedure to simulate nonisotropic scattering waves based on the 3D nonisotropic scattering theory proposed by Sato (1995). By applying the simulation method to the inversion of **M** 4.3 Wells aftershock recordings, we find that a nonisotropic scattering model, dominated by forward scattering, provides the best fit to the observed high-frequency direct *S* waves and *S*-wave coda velocity envelopes. The scattering process is governed by a Gaussian autocorrelation function, suggesting a Gaussian random heterogeneous structure for the Nevada crust. The model successfully explains the common decay of seismic coda independent of source–station locations as a result of energy leaking from multiple strong forward scattering, instead of backscattering governed by the diffusion solution at large lapse times. The model also explains the pulse-broadening effect in the high-frequency direct and early arriving *S* waves, as other studies have found, and could be very important to applications of high-frequency wave simulation in which scattering has a strong effect. We also find that regardless of its physical implications, the isotropic scattering model provides the same effective scattering coefficient and intrinsic attenuation estimates as the forward scattering model, suggesting that the isotropic scattering model is still a viable tool for the study of seismic scattering and intrinsic attenuation coefficients in the Earth.

Introduction

Over the past several decades, significant progress has been made on studies of lateral inhomogeneity of the Earth at various scales. Better seismic images and improved understanding of the composition and structure of the Earth's crust have been made through 3D seismic waveform tomography (i.e., Tape *et al.*, 2009), which have shed new light on earth dynamics and provided critical constraints for regional and global tectonic evolution. At smaller scales, however, our understanding of earth structures still relies on statistical characterization of their inhomogeneous distribution. Details of the small-scale inhomogeneities are extremely important for high-frequency wave propagation. The scattering effect of those inhomogeneities not only reduces high-frequency wave amplitudes but also redistributes their energy in space and time. Consequently, it reshapes the characteristics of the entire time series of the high-frequency velocity envelopes.

Many theories have been proposed to quantify the characteristics of the high-frequency velocity envelopes of *S* waves and *S*-wave coda. By assuming the Earth to be a random scattering medium, Aki (1969), Aki and Chouet (1975), and Sato (1977) studied high-frequency *S*-wave coda as single backscattered waves. Their work eventually led to the introduction of the radiative transfer theory in seismology (Wu, 1985) and the development of multiple scattering theories for *S* waves and *S*-wave coda velocity envelopes (Frankel and Wennerberg, 1987; Hoshiba, 1991; Zeng *et al.*, 1991; Sato, 1993). Since then, it has been a popular belief that *S*-wave codas at large lapse times are dominated by *S*-to-*S* backscattered waves. Significant progress has been made on multiple nonisotropic scattering in the last few decades. Sato (1989) used a parabolic wave equation for forward scattering to explain the pulse broadening of the direct *S*-wave group. Sato (1995) developed the theoretical framework for the calculation of nonisotropic scattering. Hoshiba (1995) studied the effect of nonisotropic scattering using a Monte Carlo simulation in western Japan focusing mostly on the spatial energy distribution in seismic coda waves. The importance of multiple nonisotropic scattering in *S*-wave coda excitation is also demonstrated by Calvet and Margerin (2013) in their observed seismic-wave envelope modeling in

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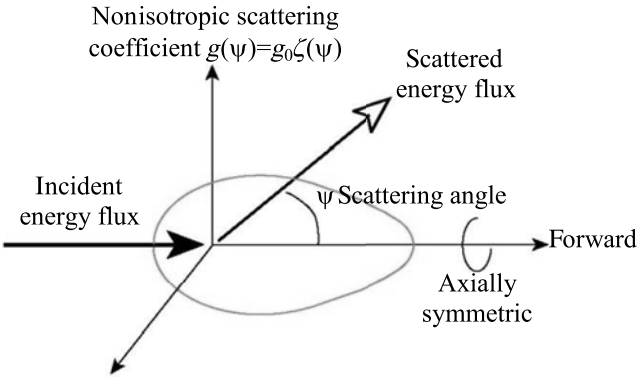


Figure 1. A schematic plot of the angular distribution of a scattering pattern $\zeta(\psi)$ as a function of the scattering angle ψ at a scatterer upon incident energy flux.

the Pyrenees. Gusev and Abubakirov (1987, 1996) calculated the nonisotropic scattering using a Monte Carlo simulation. Their result suggests possible pulse broadening for the direct-wave envelopes but shows an unrealistically fast decay in early coda after the direct wavetrain. Margerin *et al.* (2000, 2009) simulated nonisotropic *P*- and *S*-wave scattering with more realistic mode conversion to further confirm equipartition of energies between *P* and *S* waves and found equipartition occurring after a few scattering events. Sato *et al.* (2004) proposed a method to simulate the pulse broadening of the direct envelopes with increasing receiver distance using a Markov propagator and the common decay in coda-wave amplitudes with time using a 2D isotropic scattering model. An isotropic scattering model in a 2D case supports a common decay in coda-wave amplitude with time regardless of station distance but fails to explain the same phenomenon in a 3D case. Przybilla *et al.* (2009) used the elastic radiative transfer theory to successfully estimate crustal scattering parameters. The distance range covered by their data is, however, not wide enough to address any envelope pulse broadening in direct *S* waves, and the *S*-wave data could be fitted equally well using either an isotropic or a strong forward scattering model.

Despite the above-mentioned progress, most of these studies are mathematically intensive and difficult to apply. In contrast, the isotropic scattering model has been more successfully applied to observations (i.e., Wu and Aki, 1988; Fehler *et al.*, 1992; Padhy *et al.*, 2007). The same model has also become widely used in strong ground motion simulations (e.g., Zeng *et al.*, 1995; Mai *et al.*, 2010, Crempien and Archuleta, 2015; Olsen and Takedatsu, 2015) and high-frequency earthquake source imaging (e.g., Nakahara, 2008; Sawazaki and Enescu, 2014). In addition, some of the well-known phenomena commonly observed in seismic shear-wave time series have not been explained satisfactorily: (1) the common decay in *S*-wave coda independent of source–station distances (Rautian and Khalturin, 1978; Aki, 1980); (2) overall agreement between *S*-wave attenuation factor Q_S and *S*-wave coda attenuation factor Q_c (Wu, 1989); and (3) pulse broadening of the direct *S*-wave arrivals (Sato, 1989).

In this article, we build upon Jing *et al.* (2014) to study seismic scattering as part of our waveform modeling for high-frequency wave propagation. We develop an efficient numerical method to compute the scattering wave envelopes and apply it to the 2008 Wells, Nevada, earthquake aftershock recordings. By optimizing the scattering and attenuation parameters of the earth structure through inversion of the observations, we directly address the characteristics of the *S*-wave high-frequency velocity observations, such as the distance-independent decay in *S*-wave coda amplitudes, and provide physical insight to the origin of seismic scattering and useful understanding of the attenuation and heterogeneity of the Nevada crust.

Theory and Numerical Implementation

Multiple nonisotropic scattering theories have been formulated by Sato (1995) in 3D space. Following his work, we define $f(\boldsymbol{x}, t; \boldsymbol{q})$ as the directional distribution of energy density at $\boldsymbol{x}$ in the direction $\boldsymbol{q}$ at time t, in which $\boldsymbol{q}$ is a unit vector. When total energy W is spherically radiated instantaneously from the origin at $t = 0$, this directional energy density is given by

$$f(\boldsymbol{x}, t; \boldsymbol{q}) = WG(\boldsymbol{x}, t; \boldsymbol{q}) + \int_{-\infty}^{\infty}\int_{-\infty}^{\infty}\int_{-\infty}^{\infty}\int_{-\infty}^{\infty}\oint G(\boldsymbol{x}'', t - t'; \boldsymbol{q}) v g(\psi) \times f(\boldsymbol{x}', t'; \boldsymbol{q}') d\boldsymbol{x}' dt' d\Omega(\theta_q, \phi_q), \quad (1)$$

in which v is the wave velocity, and $\boldsymbol{x}''$ is the vector from the last scatterer at $\boldsymbol{x}'$ to the receiver at $\boldsymbol{x}$. The first term in the right side of equation (1) represents direct-wave propagation from the source to the receiver and is given by

$$G(\boldsymbol{x}, t; \boldsymbol{q}) = \frac{e^{-(g_0 v + b)t}}{4\pi v r^2}\delta\left(t - \frac{r}{v}\right)H(t)\delta_\Omega(\boldsymbol{x}; \boldsymbol{q}), \quad (2)$$

in which $r = |\boldsymbol{x}|$, b is the intrinsic absorption, and δ_Ω is the delta function for the solid angle Ω (see definition in A.9 and A.10 in Sato, 1995). The second term in the right side of equation (1) is given in a convolution integral that means that the energy flux in the direction $\boldsymbol{q}'$ hits the last scatterer generating scattered energy flux proportional to the scattering coefficient $g(\psi)$ in the direction $\boldsymbol{x}''$, in which ψ is the scattering angle between direction $\boldsymbol{q}'$ and $\boldsymbol{x}''$. The directional Green's function appears in the convolution integral to account for propagation from the last scatterer to the receiver. Assuming that the angular distribution of the scattered energy flux at a scatterer is axially symmetric and nonisotropic, this scattering process can be characterized by a scattering pattern function $\zeta(\psi)$ (Fig. 1) given a scattering angle ψ from the direction of the incident waves. For an isotropic scattering process, this function equals $1/4\pi$. In general,

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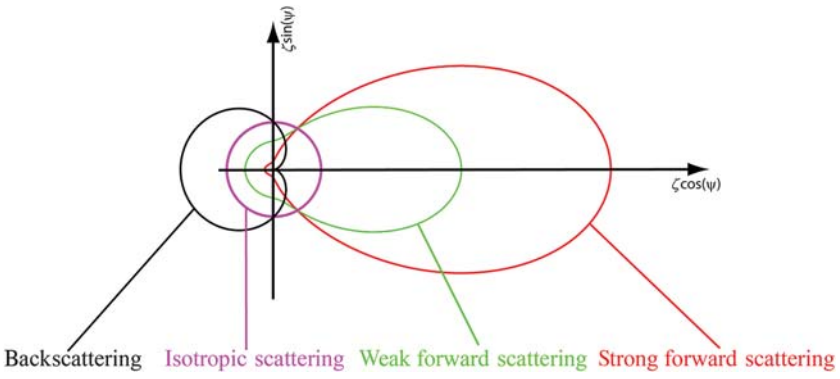


Figure 2. Scattering pattern functions for backscattering, isotropic scattering, and weak/strong forward scattering processes, respectively.

$\zeta(\psi) = g(\psi)/g_0$, in which g_0 is the total scattering coefficient. Figure 2 shows the same directional dependent scattering pattern functions but for backscattering (black), isotropic scattering (purple), and weak/strong forward scattering (green/red, respectively) processes, on the scattering plane lying along the rays of incidence and scattering. The total scattering coefficient g_0 appearing in the exponent accounts for the scattering attenuation. Integration of the directional energy density over the solid angle with respect to $\boldsymbol{q}$ gives the scattered energy density $E(\boldsymbol{x}, t)$.

The solution to equation (1) is given by Sato (1995) in the wavenumber and Laplace transfer domain. The direct and single scattering term in the solution can be calculated analytically. For higher-order scattering terms, we need an efficient numerical technique to compute the solution using Sato's formulas. These formulas include the calculation of the Gauss hypergeometrical function. We obtained a subroutine for the Gauss hypergeometrical function evaluation from the GNU online scientific library and implemented it in our numerical simulation procedure. However, the solutions became unstable when we computed higher-order terms of the function with the absolute value of the input variable $k/(i\omega + 1)$ larger than 0.5, in which k is the wavenumber, ω is the angular frequency, and i is the imaginary unit. To solve this numerical instability problem, we developed a recursive formula for the Gauss hypergeometrical function calculation based on the integral equation (15) and (36) of Sato (1995). This recursive formula is given by

$$lG_l(k_l, i\omega) = (l-1)G_{l-2}(k, i\omega) + (2l-1)\frac{i\omega + 1}{k}G_{l-1}(k, i\omega), \qquad l \geq 2, \quad (3)$$

in which $G_0 = (\tan^{-1} c)/k$, $G_1 = [1 + (\tan^{-1} c)/c]/k$, $c = k/(i\omega + 1)$, and G_l is the Gauss hypergeometrical function of order l. The same recursive relation is also given in Sato *et al.* (2012). It provides an accurate solution for $|k/(i\omega + 1)| < 0.5$ but diverges when $|k/(i\omega + 1)|$ becomes larger. Then, we replace the unstable part of the hypergeometrical function calculation with this recursive solution for $|k/(i\omega + 1)| < 0.5$. Combined with the method of discrete wavenumber sum (Bouchon, 1979), we solved the nonisotropic scattering equations numerically. The solution provides time series velocity envelopes in a nonisotropic scattering medium with uniformly distributed scatterers in a 3D space. To prepare it for seismic-wave modeling, we further modified the formula to add a free-surface response to convert a full-space model to a half-space case. This modification assumes a statistically constant free-surface correction following Hartzell and Mendoza (2011). For Sato's simple examples (Sato, 1995), our solutions agree with his corrected results (Sato *et al.*, 2012) exactly. The calculation is also validated against the analytical solution of Zeng *et al.* (1991) for isotropic scattering case.

The resulting scattered energy density $E(\boldsymbol{x}, t; \zeta, g_0, t_M)$ is given as an integration of the directional energy density in equation (1) over the solid angle, in which t_M is the parameter defining the initial source velocity impulse duration (Sato, 1989). Then, we introduce an intrinsic attenuation and a site term $A(x)$ to the scattered energy density function. The final velocity envelope function is given by the square root of the scattered energy density function as

$$\sqrt{E(x,t)} = A(x)\sqrt{E(\boldsymbol{x}, t; \zeta, g_0, t_M)}\exp(-\omega t/2\mathrm{Q}_i). \quad (4)$$

Equation (4) follows Jing *et al.* (2014) and is our basis for computing an inverse solution of the observed seismic velocity envelopes. Given the nonlinear nature of the problem, we conduct a grid search to find the optimal solution for the scattering pattern function ζ, the scattering coefficient g_0, and the source velocity impulse duration parameter t_M. Once those nonlinear parameters are determined, the attenuation parameter $b = \omega/2Q_i$ and the site term $A(x)$ can be determined with a linear least-squares method applied to the logarithm of both sides of equation (4). The model does not consider contributions from conversion scattering. It has been shown that such conversion scatterings are not significant in comparison to the *S*-to-*S*-wave scattering (e.g., Aki, 1992).

Data Analysis

Figure 3 shows the location of the Wells, Nevada, mainshock and its aftershocks. The earthquake struck northeastern Nevada on 21 February 2008. This magnitude 6.0 event was the largest earthquake to occur in Nevada in the last 30 years. During the time of the mainshock and aftershock sequences, the EarthScope Transportable Array, USArray network, was operating in Nevada. It provided a unique dataset for this earthquake sequence and a great opportunity to study wave propagation in the Great Basin area. We collected from the Incorporated Research Institutions for

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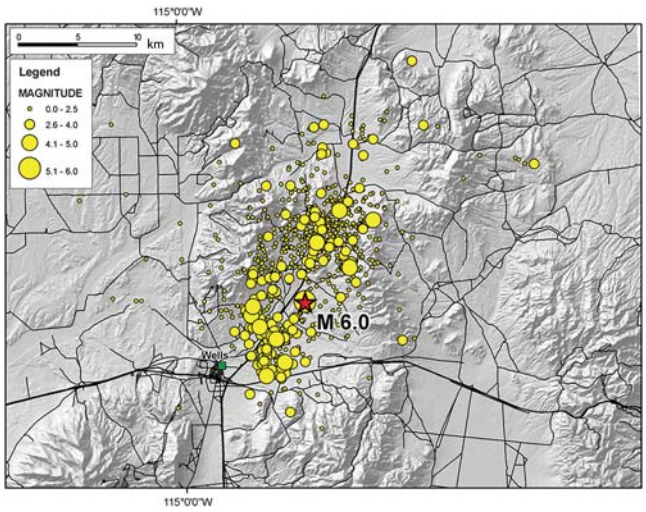


Figure 3. Map view of the February 2008 Wells, Nevada, earthquake sequence. The red star is the **M** 6.0 Wells mainshock and the yellow solid circles are the Wells aftershocks.

Seismology data center all the seismic records of this earthquake sequence with magnitude above 4.0 within an epicentral distance of 320 km of the mainshock earthquake location. Figure 4 shows the USArray station distribution and the location of an **M** 4.3 Wells, Nevada, aftershock that occurred on 22 February 2008. The distances between the epicenter and stations range from 29 to 320 km. We carefully picked the *S*-wave arrival times at all stations. Figure 5a shows examples of the vertical velocity seismograms for the **M** 4.3 event recorded at the USArray stations M12, M13, and K14, with their epicentral distances calculated at 29.5, 68.1, and 212.4 km, respectively. Our analysis in this article focuses on the vertical-component velocity seismograms. However, we applied the same modeling procedures to the other components as well and find no significant difference to the final modeling results. The data were corrected for instrument response and then band-pass filtered to calculate the velocity envelopes at each frequency. Table 1 lists the frequencies used for this study and their corresponding low and high band-pass parameters. A constant narrow bandwidth for all frequencies will result in an increasing signal pulse width with increasing frequency. Our low and high band-pass parameters were chosen to approximate the half-octave bands to best preserve the signal pulse width. We used the Seismic Analysis Code (SAC) originally developed at Lawrence Livermore National Laboratory to analyze the seismic data. Figure 5b shows the same examples as in Figure 5a but for the velocity envelope seismograms calculated using SAC with a frequency band of 9–15 Hz. Vertical red bars indicate *S*-wave arrival times. We converted all the velocity seismograms of this aftershock to velocity envelopes using SAC.

Numerical Modeling and Analysis

Before we started the inversion, we tested a range of different scattering models to understand the effect of non-

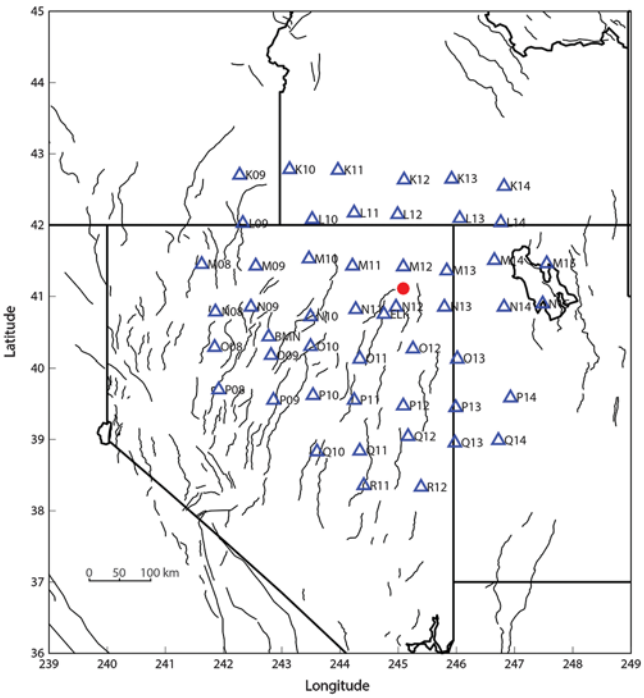


Figure 4. Map view of the aftershock location and USArray station distribution. The red solid circle is the 22 February 2008 **M** 4.3 Wells aftershock location. The blue triangles are USArray stations. The black traces are faults in the Basin and Range area.

isotropic scattering. These tests suggested that different scattering processes result in significantly different temporospatial scattering energy distributions. Figure 6 compares time series of synthetic velocity envelopes at different distances between the backscattering model (black line in Fig. 2), the isotropic scattering model (purple line in Fig. 2), and the strong forward scattering model (red line in Fig. 2) with the same scattering and intrinsic attenuation coefficient. Amplitudes of these envelopes are multiplied by their corresponding epicentral distances to plot them on the same figure. Clearly, the forward scattering model is very efficient in generating scattered waves around the direct-wave ray path. These waves follow immediately after the direct arrival to form a direct wavetrain with its total pulse width broadened as it propagates to larger distances. The backscattering model and the isotropic scattering model, on the other hand, are very efficient in generating scattered waves in directions other than the direct-wave propagation direction, resulting in redistribution of the direct-wave energies into their coda waves. In the end, they produce higher coda-wave amplitudes than that of the forward scattering model.

In Figure 7a, we plot the same examples of velocity envelopes as in Figure 5b, but with amplitudes scaled by the station epicentral distance. The purple, cyan, and green lines are the smoothed versions of the corresponding velocity envelopes calculated using the root mean squares (rms) in a 0.4-s moving window. The rms smoothed velocity envelopes have been used for data processing throughout this article. In

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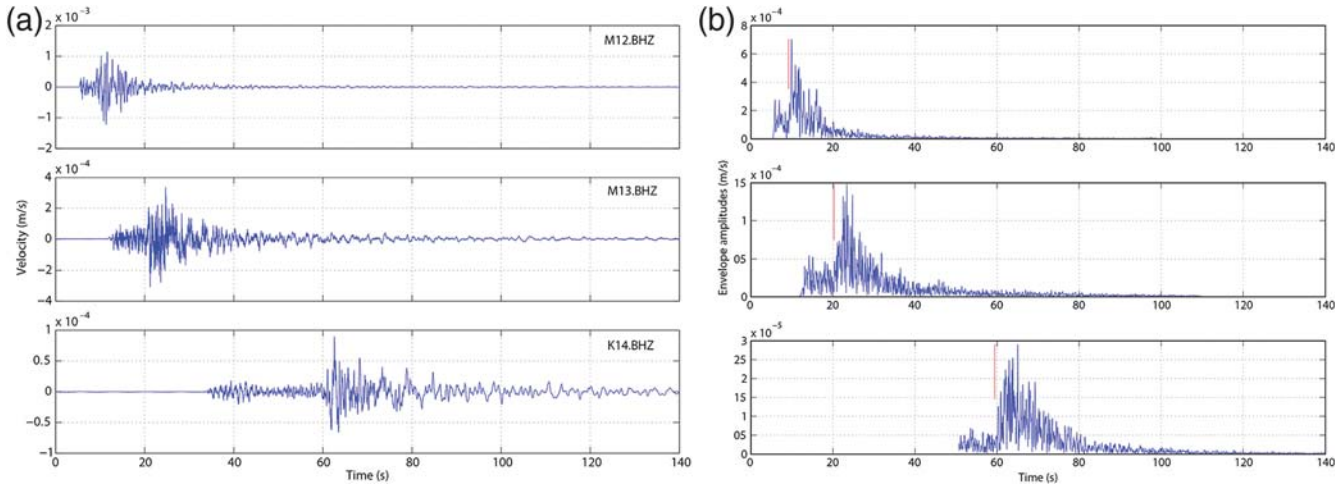


Figure 5. (a) Vertical velocity seismograms observed for the **M** 4.3 event on 22 February 2008 Wells, Nevada, earthquake aftershock at the USArray stations M12, M13, and K14, respectively. (b) Same as (a) but for the velocity envelope seismograms calculated using the Seismic Analysis Code (SAC) originally developed at Lawrence Livermore National Laboratory to analyze seismic data with a frequency band of 9–15 Hz. Vertical red bars are *S*-wave arrival times.

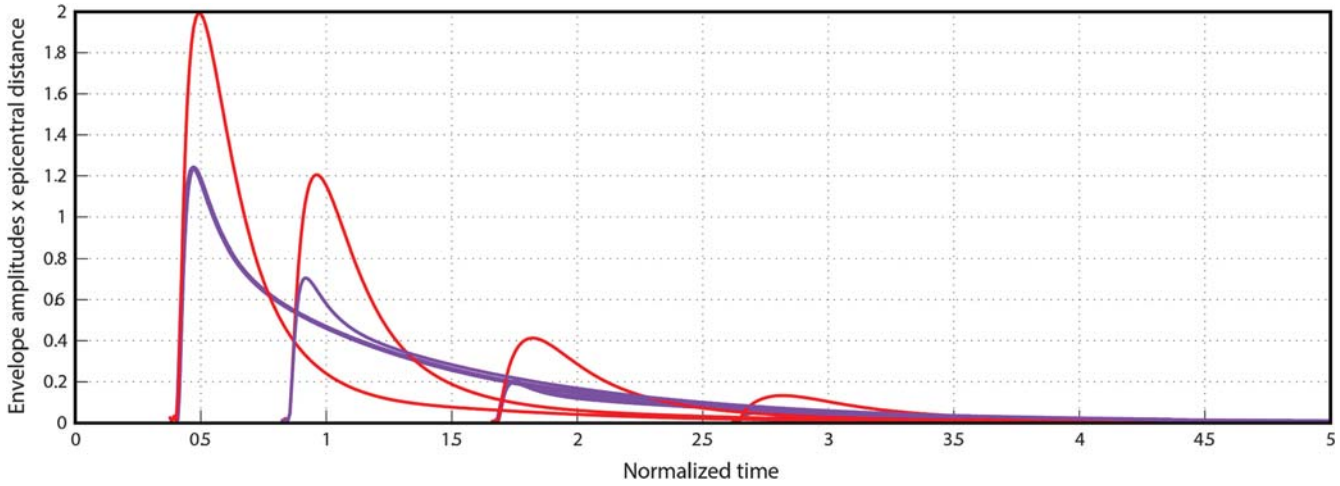


Figure 6. Comparison of time series of synthetic velocity envelopes calculated at different distances. The purple envelopes are for the isotropic scattering model (purple color in Fig. 2) and the red envelopes are for the strong forward scattering model (red color in Fig. 2).

addition, we corrected the velocity envelopes for site effects using the coda normalization method (Tsujiura, 1978) and applied it to all other envelope data used in this study. By comparing these envelopes to those in Figure 6, it is obvious that the strong forward scattering model fits better with the observations. By trial and error, we adjusted the scattering and attenuation parameters to fit the synthetic envelopes with the observed envelopes using the strong forward scattering model in Figure 6. Figure 7b plots the final comparison between observed and synthetic velocity envelopes (in black). The result strongly suggests that the forward scattering process dominates the scattering wavefield at high frequency.

To validate our solution more rigorously, we calculated a series of scattering patterns (Fig. 8, upper right corner) ranging from back scattering (black) to strong forward scattering (blue). These scattering patterns were based on the Born approximation to scattering of scalar waves in a random velocity medium characterized by a Gaussian autocorrelation function (ACF; Aki and Richards, 1980). Our test runs and a careful analysis of Jing *et al.* (2014) suggest that the scattering pattern function of a random medium characterized by a Gaussian ACF produces a better fit to the data than that of a

Table 1
List of Frequencies and Their Corresponding Band Limits Used for the Data Analysis

Frequency (Hz)	1.5	3.0	4.0	6.0	8.0	10.0	12.0	15.0
Low band limit	1.0	2.0	3.0	4.0	6.0	8.0	9.0	12.0
High band limit	2.0	4.0	5.0	8.0	10.0	12.0	15.0	18.0

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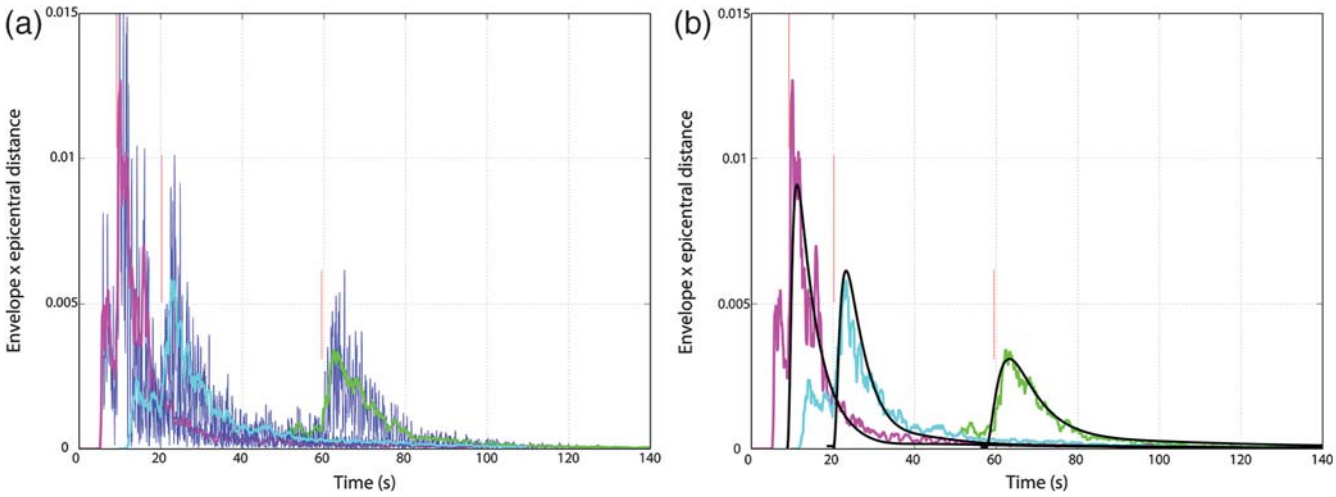


Figure 7. (a) The same velocity envelopes in Figure 5b but multiplied by the corresponding station epicentral distance to fit them in the same plot. The purple, cyan, and green lines are the corresponding smoothed envelopes calculated using the root mean squares in a 0.4-s moving window. (b) Comparison between observed (colored) and synthetic velocity envelopes (black). The observed envelopes are the smoothed version in (a). The synthetic envelopes were calculated using the strong forward scattering model in Figure 2.

random medium characterized by an exponential or a von Karman autocorrelation function. This conclusion is also supported by Hoshiba (1995). The scattering pattern function characterized by the Gaussian ACF is given by

$$\zeta(\psi) = \frac{\mu}{1-\exp(-\mu)}\exp\left(-\mu\sin^2\left(\frac{\psi}{2}\right)\right), \qquad (5)$$

in which $\mu = a^2k^2$, a is the correlation length scale of the crustal inhomogeneity, and k is the wavenumber. $\mu = 0$ and $\mu > 0$ correspond to the isotropic scattering and forward scattering models, respectively. Following a similar approach of Sato (1995), we use $\zeta_1 = -0.08$ and -0.16 to describe the backscattering model, in which ζ_1 is the first-order spherical harmonic expansion parameter of the scattering pattern function. The strongest forward scattering model corresponds to $\mu = 8$. For the other series of forward scattering models, we use linear interpolation between isotropic scattering and strong forward scattering with $\mu = 8$.

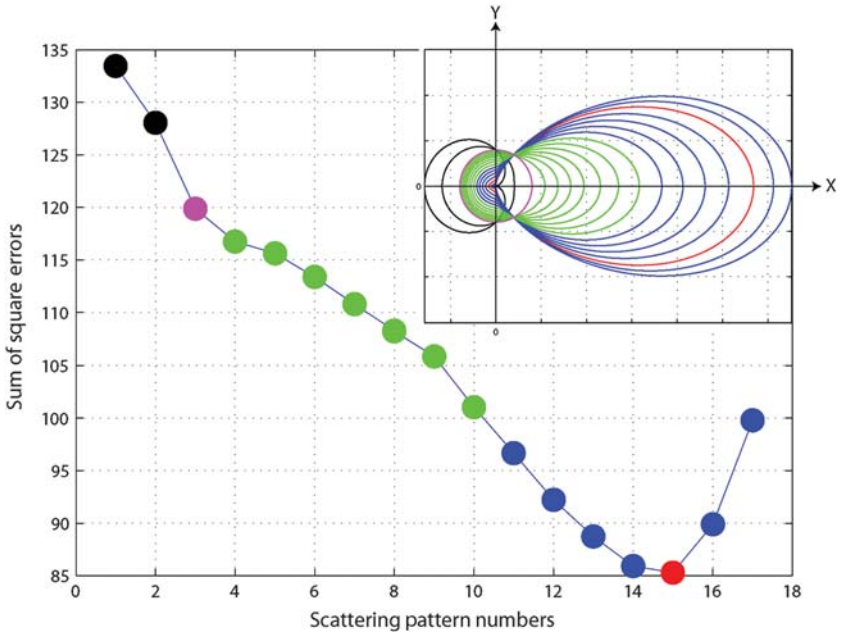


Figure 8. Sum of the square errors between synthetic and observed envelope functions versus the scattering pattern models ranging from backscattering to strong forward scattering. These model numbers are color coded in the same order as that of the scattering pattern functions plotted in the upper right corner of the figure.

Based on each scattering pattern, we then calculate synthetic velocity envelopes and determine the scattering model parameters through inversion of the data. According to equation (4), once a scattering pattern is selected, we conduct a first round of joint coarse grid search of the scattering coefficient g_0 in the 0.001–0.03 (1/km) range with a grid step of 0.001 (1/km) and the initial velocity pulse duration parameter t_M in the 0.01–0.4 s range with a grid step of 0.02 s. The intrinsic attenuation parameter is solved for using a linear least-squares inversion. Once the optimal solutions for g_0 and t_M are found, we carry out a second round of joint fine grid search in the neighborhood of the optimal $g_0 \pm 0.002$ (1/km) and $t_M \pm 0.04$ (s) with a finer grid step of 0.0001 (1/km) and 0.002 s for g_0 and t_M, respectively. Among all inversion parameters, the contribution from the initial velocity pulse duration t_M is rather insignificant and has little influence on the coda envelopes. We use this grid-search method to determine the scattering coefficient, intrinsic attenuation parameter, and the initial

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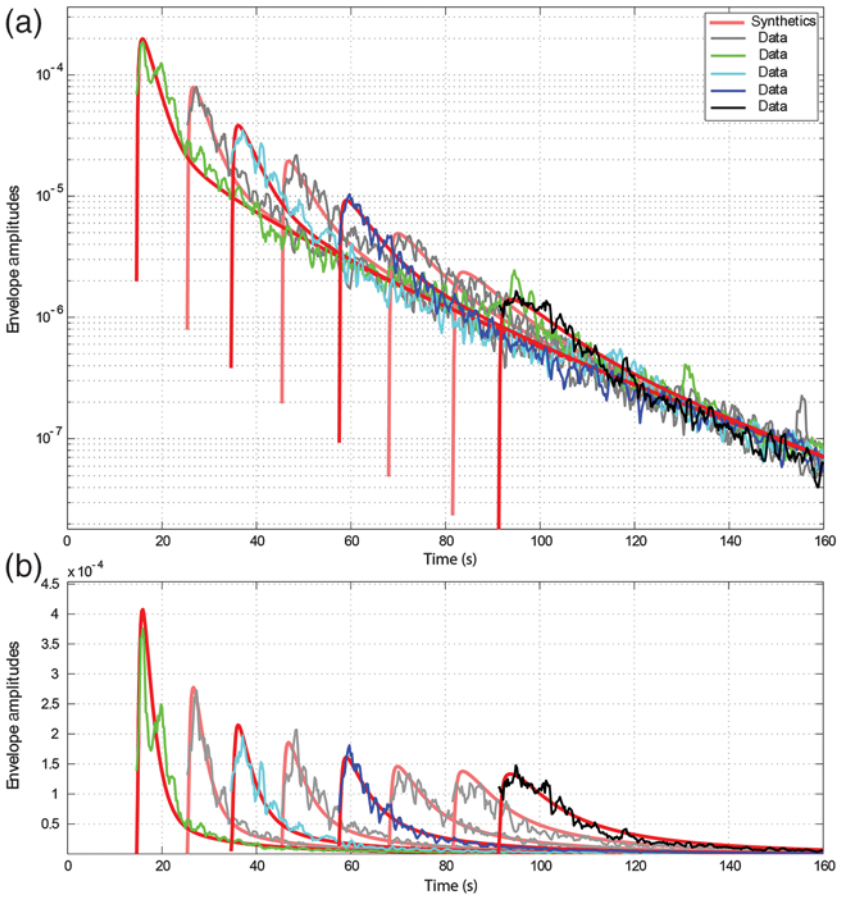


Figure 9. (a) Comparison of the synthetic velocity envelope from the best-fitting forward scattering model with the observation in logarithmic scale at several representative distances among all available stations used for the inversion at 12 Hz. The colors for different recordings are not related to the color coding in Figure 7. (b) The same comparison as in (a) but using a linear scale. Both synthetics and observations in (b) are rescaled by $\exp(ts/20)$, in which ts is the *S*-wave arrival time for the corresponding stations.

velocity pulse duration parameter from velocity envelopes of the earthquake recorded at 50 USArray stations for frequencies ranging from 1.5 to 15 Hz. The high-frequency limit is set by the data sampling rate of 0.025 s, which gives a Nyquist frequency of 20 Hz. The low-frequency limit is determined by the fit between synthetics and observations. Fits are significantly degraded for frequencies below 1.5 Hz.

Figure 8 shows the sum of the square errors between observations of the **M** 4.3 event on 22 February 2008 and synthetic envelopes as a function of the scattering pattern with a frequency band from 9 to 15 Hz. We used up to six higher-order terms to represent each scattering pattern shown in the figure. The interval of time for the error estimate starts from the *S*-wave arriving time until an average lapse time of 193 s, with the lapse time varying between 162 and 213 s. The sum of the square errors for each scattering pattern is plotted in the same color as its corresponding scattering pattern function given in the upper right corner of Figure 8. The two backscattering models (black) exhibit the worst fit to the data. As we gradually decrease the amount of backscattering and increase the amount of forward scattering, the overall scattering change from backscattering to isotropic scattering (pink), and then to increasingly forward scattering dominated processes. The sum of square errors for the isotropic scattering model is an improvement over that of the backscattering models. However, the forward scattering dominated models provide much better fits in comparison with the isotropic scattering and backscattering models. Among the forward scattering models, the error gradually decreases as the forward scattering intensity (green, blue, and red) increases. The forward scattering model shown in red provides the best fit among all other tested models. After that, any additional increases in forward scattering no longer improve the fit between synthetics and observations. Our application to the Wells aftershock recordings suggests that the traditional backscattering theory (i.e., Aki and Chouet, 1975) does not explain the observations. Instead, a forward nonisotropic scattering model provides the best fit to the observed high-frequency direct *S* waves and *S*-wave coda velocity envelopes.

Examining the temporal behaviors in the velocity envelopes between the synthetics and observations is instructive. Figure 9 shows the comparison of the synthetic velocity envelopes based on the best forward scattering model (the model in red in Fig. 8) with observations at 12 Hz at several representative distances among all available stations used for the inversion. The observed envelope functions recorded at different distances are indicated by their *S*-wave onset times. To view both direct *S* waves and *S*-wave coda more clearly, we plot these envelopes on a logarithmic scale in Figure 9a. The synthetic and observed envelopes agree very well for both direct *S* waves and *S*-wave coda at all distances. We also observe a common decay curve in coda independent of station epicentral distance for both synthetics and observations. This phenomenon was first found from observations made in the 1970s and early 1980s (i.e., Rautian and Khalturin, 1978; Aki, 1980) and was used for coda normalization (Tsujiura, 1978; Aki, 1980) and for estimation of site amplification (i.e., Phillips and Aki, 1986; Su *et al.*, 1992). In our synthetic test, we found that the common decay observed in seismic coda is apparently a direct consequence of multiple forward scattering. Both backscattering and isotropic scattering models generate coda waves at different distances that separate from each other in amplitude at later lapse times within the time frame of this study, although we recognize that they will eventually converge at times that are

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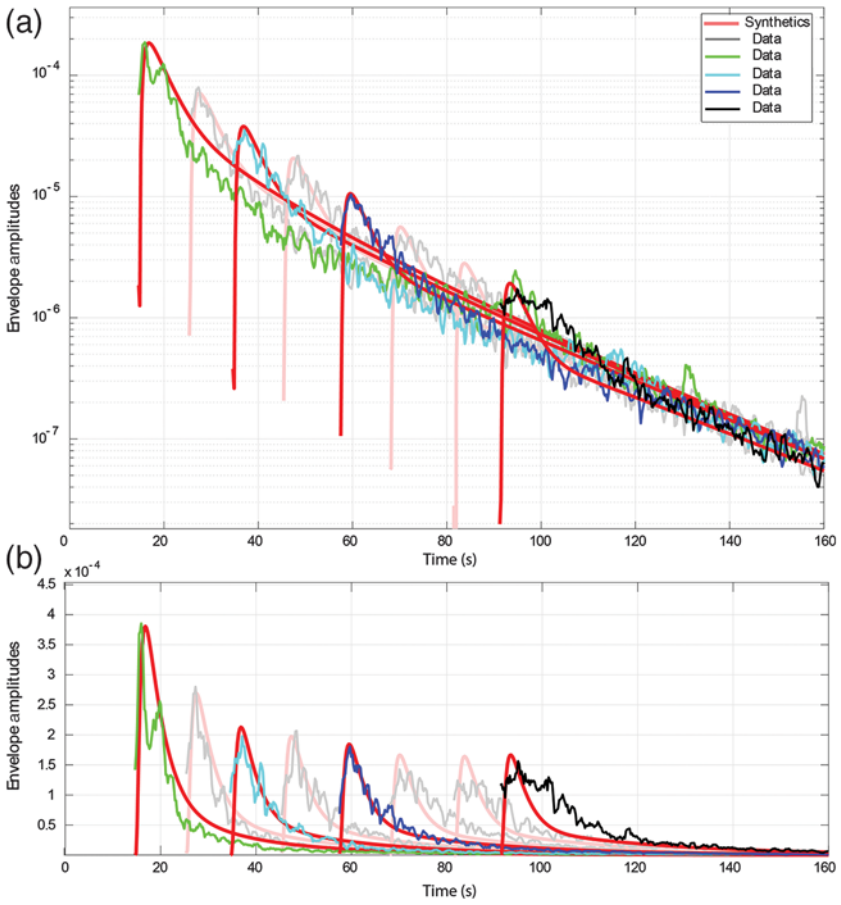


Figure 10. (a) Comparison of the synthetic velocity envelope from a best-fitting backscattering model with the observation in logarithmic scale at several representative distances among all available stations used for the inversion at 12 Hz. (b) The same comparison as in (a) but in linear scale. Both synthetics and observations in (b) are rescaled by $\exp(ts/20)$, in which ts is the S-wave arrival time for the corresponding stations.

much larger than the transport mean free time (Ishimaru, 1978).

Another important consequence of forward scattering is envelope broadening. To examine the effect of envelope broadening on direct S waves and early S-wave codas, we plot both synthetic and observed envelopes on a linear scale (Fig. 9b). We scale each envelope function by a factor of $\exp(ts/20)$, in which ts is the corresponding S-wave arrival time at the station. Again, we see the same agreement between synthetic and observed velocity envelopes. We also find that the envelope pulse widths of the direct S-wave velocities and their early codas increase as distance increases. Because the duration of the pulse width and the consequential broadening are much longer than that of the source duration, the broadening is clearly a result of scattering wave propagation. Our synthetic analysis demonstrates that the pulse broadening effect is caused by a combined result of a decrease in amplitude of the direct S wave and an increase in amplitude of the early S-wave codas as a result of an accumulative multiple forward scattering. This envelope pulse broadening effect is commonly observed in high-frequency wave propagation at regional distances. The best-fitting forward scattering model is characterized by a Gaussian ACF with $\mu = 7.2$.

Figure 10 plots the same comparison as Figure 9 of the synthetic velocity envelopes with observations at 12 Hz at several representative distances but for the best-fit backscattering model (the model shown in black in Fig. 8). As we would expect, the coda-wave velocity envelopes generated from the backscattering model show a distinct departure from their observed envelopes at large lapse times (Fig. 10a). The inversion produces a good fit between the synthetic velocity envelopes and the observations at middle distance ranges. But as a compromise, it overestimates the observed coda waves at near distances and underestimates coda waves at larger distances. The backscattering model also propagates direct S-wave groups with a constant pulse width in contrast to the observed pulse width broadening in the data (Fig. 10b). Again, the inversion overestimates the total pulse width of the direct S-wave groups at near distances and underestimates them at larger distances.

For all other frequencies, we plot the best fitness function relative to that of the backscattering model (Fig. 11). The fitness functions are defined by the same sum of the square errors between observed and synthetic envelopes. The fit degrades as the frequency decreases, suggesting that the component of backscattered waves increases at lower frequencies. Based on F-test statistics, we found that the differences in goodness of fit between the best-fit strong forward scattering models and the isotropic/backscattering models are significant over the 95% confidence level for all frequencies except at 1.5 Hz. In addition to the increase in backscattering wave amplitudes, surface-wave scattering amplitudes also increase at frequencies below 1 Hz. In this study, our scattering models apply only to the seismic body-wave propagation and scattering.

Scattering and Intrinsic Attenuation

As seismic waves propagate through the Earth, their amplitudes decrease with increasing distance partly because of the effect of geometrical spreading and most importantly because of the scattering and intrinsic attenuation. Figure 12 plots the scattering attenuation factor Q_{sc} (blue) and intrinsic absorption factor Q_i (red) as a function of frequency from 1.5 to 15 Hz for the optimal model of the inversion. Table 2 lists

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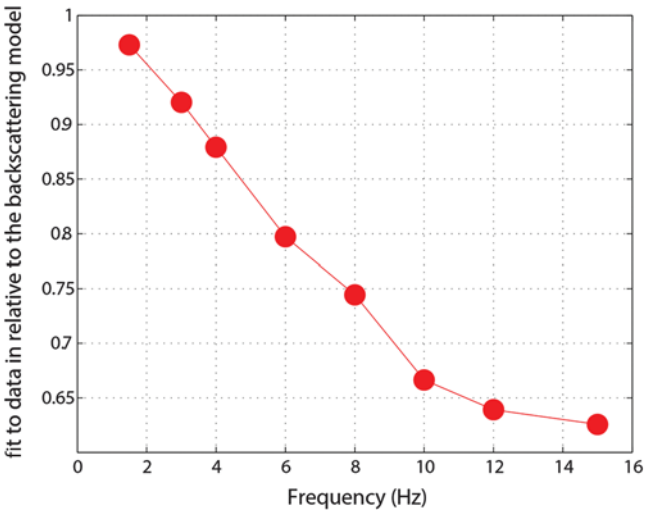


Figure 11. Goodness of fit for the best-fitting forward scattering model relative to the worst backscattering model at each frequency from 1 to 15 Hz.

the results of the scattering coefficient g_0, scattering attenuation factor Q_{sc}, and intrinsic attenuation factor Q_i at all frequencies, and the direct *S*-wave attenuation factor Q_s estimated from the observed direct *S*-wave decay with distance. It is clear that the scattering attenuation $1/Q_{sc}$ shown in blue decreases with increasing frequency, suggesting $ak > 1$, in which a is the length scale of the crustal inhomogeneity, $k = 2\pi/\lambda$ is the wavenumber, and λ is the wavelength. This result indicates that the wavelengths used in this study are much shorter than or comparable with the typical length scale of the crustal inhomogeneity in the area. For a random inhomogeneous elastic medium characterized by a Gaussian or an exponential autocorrelation function for its seismic velocity distribution, it was found that scattering shows a strong forward scattering pattern at high frequencies, $ak > 1$ (Sato *et al.*, 2012), that is, with a large scattering amplitude in a narrow cone around the forward direction defined by $\psi < 1/ak$. Given our best scattering model as a Gaussian random media with $\mu = a^2k^2$ varying from 5.8 to 7.2 for a frequency range from 1.5 to 15 Hz, we can estimate the crustal inhomogeneity correlation distance a in the 0.9–0.1 km range assuming an *S*-wave velocity of 3.5 km/s. Considering the uncertainty of these estimates, for example, at 1.5 Hz, a confidence interval over 68% gives the correlation distance a ranging from 0.01 to 0.95 km with the best

Table 2
List of Frequencies, g_0, Q_{sc}, Q_i, and Q_s for the Optimal Model

Frequency (Hz)	1.5	3.0	4.0	6.0	8.0	10.0	12.0	15.0
g_0 (1/km)	0.0195	0.0142	0.0111	0.0110	0.0095	0.0084	0.0088	0.0077
Q_{sc}	69.0	189.0	323.5	490.7	751.9	1073.4	1218.8	1760.0
Q_i	259.6	397.6	487.1	684.0	844.8	1055.4	1341.8	1740.3
Q_s	133.8	260.6	353.9	507.2	659.6	847.1	1052.2	1395.4

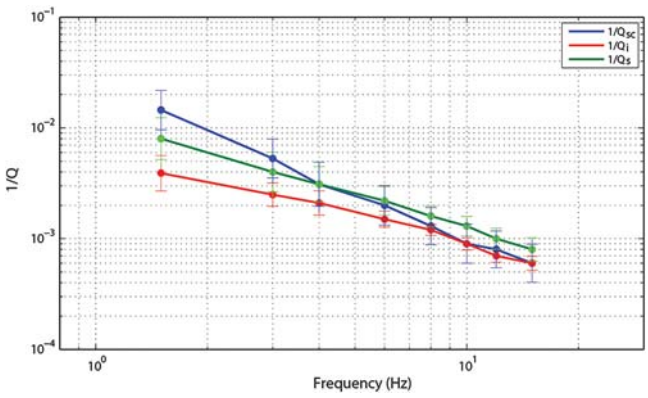


Figure 12. Scattering attenuation factor Q_{sc} (blue), intrinsic absorption factor Q_i (red), and direct *S*-wave attenuation factor $1/Q_s$ (green) versus frequency for the Nevada region. Vertical bars are the corresponding errors for the measurements.

estimate $a = 0.90$ km, based on *F*-test and chi-square distributions as a function of the scattering parameters. The lower and upper error bars are nonsymmetrical for 1.5 Hz. At 15 Hz, however, a confidence interval over 68% gives the correlation distance a ranging from 0.09 to 0.11 km. Giving the uncertainty of the correlation distance estimates, it is possible for a constant $a = 0.10 \pm 0.01$ to fit to all frequencies used in this study. The best estimates, however, suggest a set of correlation distance a, particularly at lower frequencies.

The intrinsic attenuation $1/Q_i$ (red) in Figure 12 also decreases with frequency, similar to $1/Q_{sc}$ at higher frequency, but gradually flattens toward lower frequency. Part of the apparent frequency dependence in $1/Q_i$ could be caused by the energy leaking to the lower crust and upper mantle (Margerin *et al.*, 1999). Intrinsic attenuation reduces both the direct *S*-wave amplitude and the *S*-wave coda amplitude with propagation distance and time, whereas scattering reduces the direct-wave amplitude but increases the coda wave amplitude. The overall quality factor for *S* waves is given by

$$1/Q_s = 1/Q_i + 1/Q_{sc}. \quad (6)$$

To compare the *S*-wave attenuation to the *S*-wave coda attenuation, we measured both the direct *S*-wave attenuation factor Q_s from *S*-wave amplitude decay with distance and the *S*-wave coda Q_c from the coda decay with time. The direct *S*-wave attenuation factor Q_s was measured using the method given in box 5.7 of Aki and Richards (2002). The coda Q_c was measured using the method of Aki and Chouet (1975). The value of Q_c is approximately equal to the intrinsic attenuation factor Q_i. The direct *S*-wave attenuation factor Q_s is plotted in green solid circles in Figure 12. Clearly, the inverse of the

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measured *S*-wave attenuation factor Q_s does not follow equation (6) as a sum of the intrinsic and scattering attenuations. Instead, we find that

$$1/Q_s = 1/Q_i + \alpha/Q_{sc}, \tag{7}$$

in which α is a fraction measured to be around 0.3 for all frequencies used in this analysis. If we define α/Q_{sc} as the effective scattering attenuation, we can argue that the relationship between the effective scattering attenuation and the actual scattering attenuation follows the same linear relationship between the effective scattering mean free path and the actual scattering mean free path of the medium. Following Ishimaru (1978), we can calculate the effective mean free path for the scattering diffusion process at large scattering times. The effective mean free path is related to the actual scattering mean free path by a factor of

$$\alpha = 1 - \langle \cos\psi \rangle, \tag{8}$$

in which $\langle \cos\psi \rangle$ is the mean cosine of the scattering angle ψ weighted by the scattering pattern function $\zeta(\psi)$ (Fig. 1) over all possible values of ψ from the direction of the incident waves. In our case, this factor is calculated at 0.33, approximately equal to the factor α that we measured in the relation between the effective scattering attenuation and the total scattering attenuation $1/Q_{sc}$. This effective mean free path is also called the transport mean free path. The estimate of effective scattering loss is very similar to the travel-time-corrected scattering loss (Sato, 1982) and scattering loss into the backward half-space (Wu, 1982). Because the effective scattering attenuation is only a small fraction of the total scattering attenuation, it explains why there is a close agreement between *S*-wave attenuation and *S*-wave coda attenuation.

Analysis of the Scattering Coefficients

In addition to the effective scattering attenuation parameters, we also calculated the effective scattering coefficient g_0 for the optimal strong forward scattering model (red in Fig. 8) and the optimal isotropic scattering model (purple in Fig. 8). We define the effective scattering coefficient g_0 as g^*. The relationship between g_0 and g^* is the same as that between the scattering attenuation coefficient and the effective scattering attenuation coefficient. Our estimate gives a g^* of 0.0029 1/km at 12 Hz for both the forward scattering and isotropic scattering models. For other optimal models ranging from backscattering to weak and strong forward scattering models, g^* equals 0.0029 ± 0.0004 1/km. This result suggests that g^* estimate is independent of the scattering model. To confirm this conclusion for all other frequencies, we repeated the same calculation for all scattering models. We found that even though it is frequency dependent, g^* is independent of the scattering model within the error of the estimates. The result shows that the isotropic scattering model provides the same effective scattering parameter estimates as other more sophisticated forward scattering models. This conclusion suggests that the isotropic scattering model is still a viable model for estimating scattering and intrinsic attenuation parameters, for example, the multiple lapse-time window analysis based on the isotropic scattering model (i.e., Fehler *et al.*, 1992, Mayeda *et al.*, 1992; Sato *et al.*, 2012). The results from seismic scattering and intrinsic attenuation studies over the past several decades using the isotropic scattering model are still valid despite our progress made today in understanding the seismic scattering processes of the Earth.

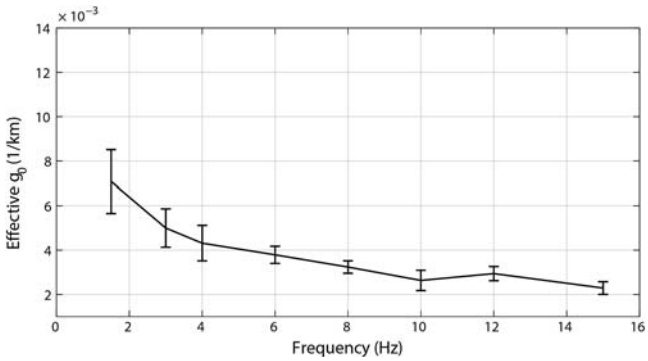


Figure 13. The effective scattering coefficient g_0 estimate versus frequency.

Przybilla *et al.* (2009) have shown that the relationship between transport mean free path and wavenumber depends on the characteristics of the random scattering medium. Based on their result, we can easily obtain that g^* is proportional to $\ln(f)$ and $f^{(1-2\kappa)}$, with $0 < \kappa < 0.5$, for an exponential medium and a von Karman medium, respectively, in which f is the frequency. This derivation means that g^* increases with frequency either logarithmically or as a power function of frequency if the medium is an exponential or a von Karman random medium, respectively. Figure 13 plots our g^* as a function of frequency. We find, however, that the estimated g^* decreases with increasing frequency and gradually reaches a constant value at high frequencies. This result indicates that the structure underneath Nevada is neither an exponential nor a von Karman type medium, suggesting that the assumption of an exponential medium for the Nevada crust in Jing *et al.* (2014) is not a valid approximation. For a Gaussian random medium, g^* approaches a constant value at high frequency (Przybilla *et al.*, 2009), agreeing with the behavior of g^* shown in Figure 13. This result supports our initial assumption of a Gaussian random medium for Nevada.

Discussion

We applied a combined grid search and linear inversion method to invert for the scattering coefficients and intrinsic attenuation parameters from velocity envelopes of a 2008 **M** 4.3 Wells, Nevada, aftershock for frequencies ranging from 1.5 to 15 Hz. The results suggest that a strong forward scattering model (Figs. 8 and 11) best fits the data at all

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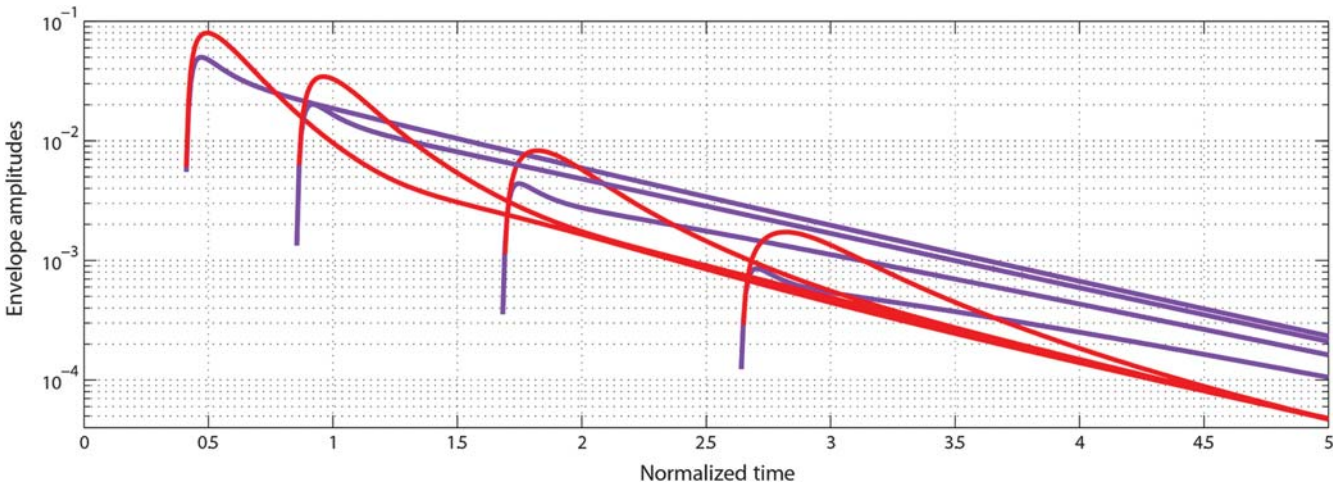


Figure 14. The same comparison of time series between synthetic velocity envelopes calculated at different distances as in Figure 6 but plotted in a logarithmic scale. The purple envelopes are for the isotropic scattering model and the red envelopes are for the strong forward scattering model.

frequencies. As frequency decreases, the fit gradually degrades. After carefully examining the waveforms in the seismograms, we found that surface waves gradually dominate the wavefield at larger distances for frequencies below 1 Hz. The degrading fit is partly a result of a gradual increase in backscattering and an increased presence of surface-wave scattering. Our current scattering theory applies only to the seismic body waves.

In Figure 14, we replot the same comparison of the time series of synthetic velocity envelopes at different distances as in Figure 6 but on a logarithmic amplitude scale. The difference in late coda waves between different scattering models becomes quite pronounced. In contrast to the strong forward scattering model showing a common energy decay in late-arriving coda waves, the isotropic scattering model shows departing asymptotic decays in late codas between stations at different distances at large lapse time. In fact, if we define our fitness function as the sum of square errors between the logarithm of the observed and synthetic envelopes, the difference in the goodness of fit would be more significant between the forward scattering models and the backscattering models.

The forward scattering model provides a theoretical explanation to both the pulse broadening of the direct *S*-wave group and a common energy decay in the velocity envelopes of seismic coda regardless of the source–station distances. By carefully identifying the direct waves and the scattering waves in the numerical simulation, we find that the group of waves that appears in the seismograms as the direct waves consists of not only the direct arrival but also a collection of forward scattered waves. For a short travel distance, the direct arrival dominates this group of waves, resulting in a narrower velocity envelope pulse determined mainly by the rupture duration of the source. As source–station distance increases, the amplitude of the direct-wave attenuates and the forward scattered waves gradually take over. Those forward scattered waves travel around the direct wavepath and disperse gradually as their travel distance increases, resulting in a broadening of their collective velocity envelope pulse.

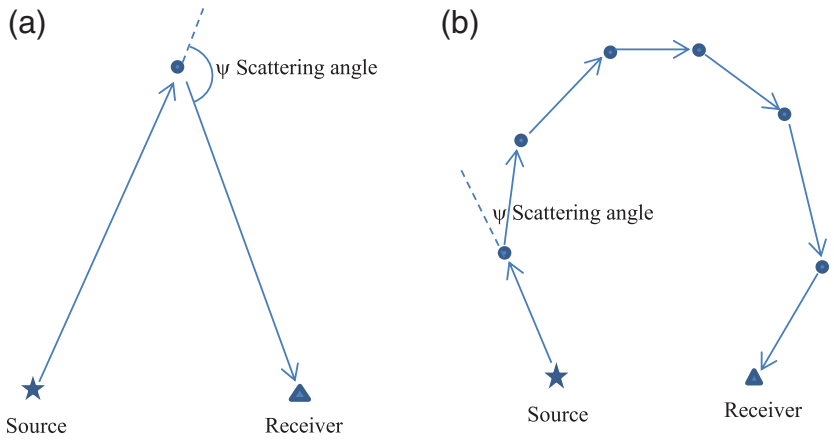


Figure 15. Schematic plot of (a) a single scattering ray path and (b) a multiple scattering ray path from source to scatterers, and then to the receivers, respectively.

Figure 15a shows a ray path of a single backscattered wave from source to a scatterer, and then to the receiver. For scattering distances larger than the source–receiver distance, the incident waves are scattered with a large scattering angle into a backscattering direction to the receiver. For the isotropic scattering case, the amplitude of the scattered waves is independent of their scattering angle. Consequently, the collection of these single

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scattered waves in seismic coda follows a common decay regardless of epicentral distance, providing a convenient explanation for seismic coda observations (Aki and Chouet, 1975). However, as time increases, the multiple scattered waves to the same receiver gradually dominate the seismic coda, and their energy distribution does not follow a common energy decay curve (Zeng, 1991). Assuming a strong seismic scattering presence at a shallow depth, we could simulate the coda velocity envelope decay asymptotic to a common decay curve (Zeng, 1991; Hoshiba, 1994), but we could not simulate the envelope pulse broadening for the direct-wave group with the isotropic scattering mechanism.

In a strong forward scattering medium, the backscattered waves are negligible in their amplitude compared with the amplitude of forward scattered waves because of their significantly lower scattering amplitudes as a function of the scattering angle (Fig. 2). As a result, scattered waves that dominate the seismic coda at late times are waves scattered in a small scattering angle multiple times in space. Figure 15b shows an example of a scattered wave ray path from the source to the receiver but through a multiply accumulative forward scattering process. By eliminating a wide range of combinations of multiply scattered waves having large scattering angles, this forward scattering mechanism allows only certain types of scattering waves similar to those shown in Figure 15b to contribute to the seismic coda. These multiple forward scattering contributions to the late seismic coda follow a common energy decay curve in much the same way as the single isotropic scattering waves do mathematically, but with a very different scattering mechanism. They follow a very similar collection of ray paths as that of the single scattered waves but are scattered multiple times along their way and form a more rounded shape in their actual ray path.

It is well known that the coda envelope at large lapse times is governed by the diffusion solution with the effective transport scattering coefficient, which neglects forward scattering but enriches backscattering by a factor of $1-\cos\psi$ (Sato *et al.*, 2004, 2012). To compare our strong forward model result with the diffusion solution, we plot a comparison of the synthetic velocity envelopes from the optimal forward scattering model at 12 Hz with that of a diffusion model (green) governed by the transport scattering coefficient using a factor of $\langle 1-\cos\psi\rangle$ in Figure 16. Following Sato *et al.* (2012), the diffusion solution is given by

$$E^D(x,t)=\frac{W}{(4\pi D_c t)^{3/2}}\exp\left(-\frac{r^2}{4D_c t}-\omega t/\mathrm{Q_i}\right), \qquad (9)$$

in which $D_c = V/(3g_0\langle 1-\cos\psi\rangle)$, V is the S-wave velocity, W is the source energy, and $\mathrm{Q_i}$ is the intrinsic attenuation factor. We replaced the original scattering coefficient g_0 with the effective scattering coefficient $g_0\langle 1-\cos\psi\rangle$. The source energy amplitude of the diffusion solution is scaled to the source energy amplitude of the synthetic forward scattering solution. The diffusion solutions at different distances do not follow a common decay curve at large lapse times within the time window considered here. For additional comparison, we plot the backscattering solutions from Figure 10a (in black). Although the diffusion solutions compare more favorably with the strong forward scattering results at early lapse times, they agree better with the backscattering solutions at late lapse times. Thus, for the strong forward scattering case, our result does not support the conclusion that coda waves at large lapse times within the scope of this study are governed by the diffusion process defined by the effective scattering coefficient. However, they will eventually converge at times that are much larger than the transport mean free time (Ishimaru, 1978).

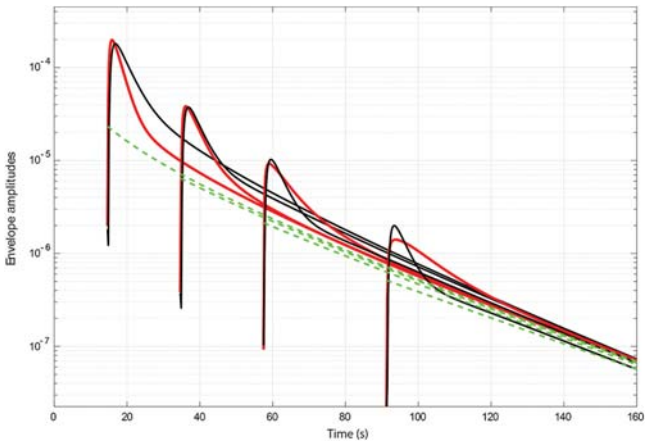


Figure 16. Comparison of the synthetic velocity envelopes from the optimal forward scattering model (red) at 12 Hz with that from the backscattering model (black in Fig. 2) and from a diffusion model (green) governed by the transport scattering coefficient using a factor of $1-\langle\cos\psi\rangle$. The diffusion solutions are scaled to the synthetic forward scattering solution at the source.

Another concern of our scattering modeling is whether the Born approximation used to derive the scattering pattern given by equation (5) is satisfied in our scattering model estimation. Usually the Born approximation is derived under the condition of $\varepsilon \ll 1$ and $ak \ll 1$, in which ε is the rms of the fractional fluctuation of S-wave velocity. At high frequencies (i.e., $ak > 1$), however, $\varepsilon ak \ll 1$ is required to fulfill the Born approximation (Sato *et al.*, 2012). For a Gaussian ACF, we have

$$g_0 = (\varepsilon k)^2 a(1-e^{-(ak)^2})\pi^{1/2} = (\varepsilon ak)^2(1-e^{-(ak)^2})\pi^{1/2}/a \qquad (10)$$

(Sato *et al.*, 2012). Giving $(ak)^2 \approx 5.8$–7.2, we find $(1-e^{-(ak)^2}) \approx 1.0$. We also find that $a \approx 0.1$–0.9 and $g_0 \approx 0.001$. Substituting these numbers into equation (10), we get

$$0.001 \approx (\varepsilon ak)^2\pi^{1/2}/a. \qquad (11)$$

Solving (εak), we find

$$\varepsilon ak \approx \sqrt{0.001a/\pi} = 0.018\sqrt{a} \ll 1. \qquad (12)$$

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So the model estimation strictly fulfills the condition for the Born approximation.

The success of the current scattering model seems to suggest that scatterers are distributed uniformly in space. In a real Earth, the scatterers are more likely concentrated at shallow depths based on many high-precision seismic tomography studies (e.g., Tape *et al.*, 2009). This structure could be compromised, however, when strong reflection boundaries exist immediately below the densely distributed scatterers. Such boundaries act like mirrors, allowing scatterers present in shallow depths to be mirrored into deeper space. Consequently our half-space model with scatterers distributed uniformly in space provides a viable alternative to real Earth models that trap waves in the upper-crustal layers. The reduction in wave amplitude following each reflection adds to the intrinsic attenuation loss. This additional reflection attenuation to the intrinsic attenuation seems to explain in part the observed decrease in intrinsic attenuation with frequency in Figure 12. Clearly, further investigation to separate the reflection attenuation from intrinsic attenuation is needed.

To test the effect of the source-to-station distance, we select stations within a distance of 120 km from the source. This distance range is similar to that used by Przybilla *et al.* (2009). We then recompute the inverse solution. Figure 17 plots the sum of square errors for the inversion with a recording distance range of 120 km in comparison with the sum of square errors for the inversion of the entire recording distance range of 320 km from the **M** 4.3 event. The sum of square errors for the distance range of 120 km is plotted below that for the distance range of 320 km. It is clear that a shorter distance range results in a flat goodness-of-fit curve as a function of scattering pattern functions ranging from backscattering to strong forward scattering. Although the result still gives a slight edge to the strong forward scattering model, overall, the distance covered by the data is insufficient to produce a model with significantly better fit than that of other scattering models. This suggests that for the recording distance range of Przybilla *et al.* (2009), an isotropic scattering model could fit nearly as well as that of a strong forward scattering model. This is also supported by Gaebler *et al.* (2015) whose study found that scattering models with a wide range of inhomogeneity scales fit the data equally well for distances up to 140 km. The forward scattering model applies well to the data to a distance range up to 320 km. We examined the *Sg* and *Sn* phases and found that the separation of *Sg* and *Sn* phases does not contribute to the envelope broadening because *Sn* phase in this region is weakly visible. This is also evident by Buehler and Shearer (2014). The broad difference in direct *S* waves and late coda only becomes apparent as distance increases. With data covering a wide range of recording distances in Nevada from the USArray project and rigorous fits to the observations, we are able to prove that the 3D nonisotropic scattering model dominated by strong forward scattering provides a unified answer to all three well-known phenomena in seismic shear-wave observations: (1) the common decay in the velocity envelopes of seismic coda regardless of the source–station distances, (2) a close agreement between *S*-wave attenuation and seismic coda-wave attenuation, and (3) pulse broadening of the direct *S*-wave group.

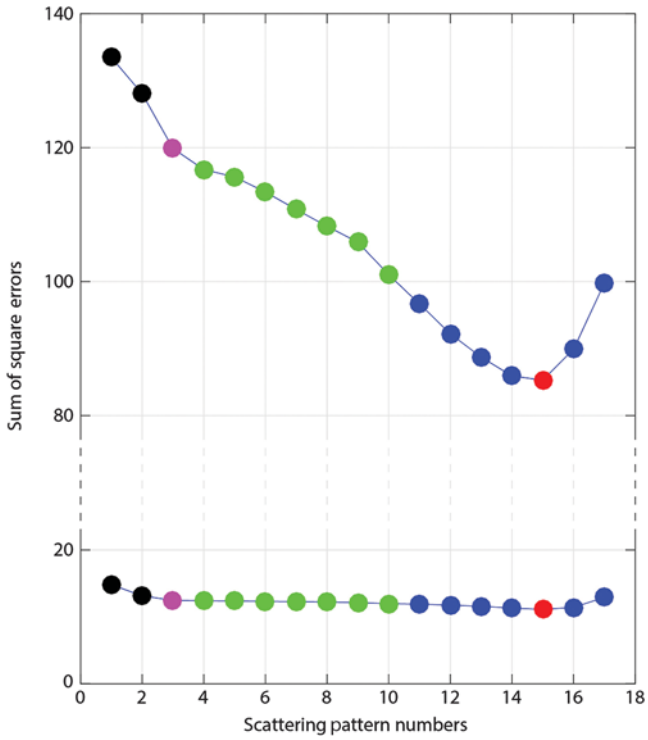


Figure 17. Comparison of the sums of square errors between synthetic and observed envelope functions as a function of different scattering models plotted in Figure 8 over different recording distances. The lower curve is for recording distances that range from 0 to 120 km and the upper curve is for the distances of 0 to 320 km as in Figure 8.

Conclusion

This article is a pilot study of the nonisotropic scattering process based on radiative transfer theory and its application to the observation of an **M** 4.3 aftershock recording from the 2008 Wells earthquake sequence in Nevada. The results show that a forward scattering model provides the best fit to the recorded high-frequency velocity envelopes of the Wells aftershock. This study provides a unified answer based on the strong forward scattering process to the decades-long observations of direct seismic shear-wave energy pulse broadening, the common energy decay in late seismic coda regardless of the source–station distances, and the close agreement between *S* wave and its coda-wave attenuation. Although these are well-known phenomena, a rigorous proof of the result based on radiative transfer theory has only become possible through USArray observations, which have provided a wide range of distances from the earthquake source to the recording stations and made it possible to

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discriminate different scattering models with distance-dependent behaviors. The model will have important applications in high-frequency wave modeling in which scattering has a strong effect.

The results of this article also show that the estimates of the effective scattering coefficient g_0 and the effective scattering attenuation $1/Q_{sc}$ are independent of the actual scattering model used. Furthermore, the results suggest that the isotropic scattering model is still a viable tool for estimating scattering and intrinsic attenuation from observations. The results from seismic scattering and intrinsic attenuation studies over the past several decades based on the isotropic scattering model (e.g., Wu and Aki, 1988; Fehler *et al.*, 1992, Mayeda *et al.*, 1992; Sato *et al.*, 2012; Padhy *et al.*, 2007) are still applicable today despite our improved understanding of the scattering process in the Earth.

This study also found that simulations dominated by strong forward scattering in a 3D random medium do not support the well-known theory that the coda envelope at large lapse times is governed by the diffusion solution with the transport scattering coefficient within the time window of this study, although we recognize that they will eventually converge at times much larger than the transport mean free time. The late seismic codas are scattered wave energy leaking out of the forward scattering process. Diffusion is a closer approximation to the backscattering process at large lapse times. Finally, this study found that the estimated effective scattering coefficient in Nevada fits the characteristics of a Gaussian ACF, suggesting a Gaussian random medium for the inhomogeneous structure in the region.

Data and Resources

Seismograms of the 21 February 2008 Wells earthquake aftershocks used for this study are recorded by the EarthScope USArray network in Nevada. All the data were collected from the Incorporated Research Institutions for Seismology (IRIS) data center.

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